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Bayesian fused lasso regression modeling for binary data via horseshoe prior

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1 Introduction

L_1-norm regularization is the popular tool for analyzing data with sparse structure. Tibshirani et al. (2005) proposed fused lasso by imposing the sum of the absolute values (L_1-norms) of regression coefficients and their successive differences as a penalty term on the log-likelihood function. By this L_1-norm penalty, fused lasso enables variable selection and variable fusion. In Bayesian method, Kyung et al. (2010) proposed Bayesian fused lasso assuming a Laplace prior on regression coefficients and their successive differences.

However, a Laplace prior tends to lead to the over-shrinkage of regression coefficients which should not be zero. To overcome this problem, horseshoe prior (Carvalho et al., 2010) is often utilized. In this paper, we propose the logistic regression model with Bayesian fused lasso modeling with horseshoe prior.

2 Logistic regression model with Bayesian fused lasso

Suppose that we have the binary response variables $\boldsymbol{y} = (y_1, \ldots, y_n)^T \in \{0, 1\}$ and p-dimensional vector of explanatory variables $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$. Logistic regression model is formulated as

$$\Pr(y_i = 1 \mid \boldsymbol{x}_i) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta})}},\ i = 1, \ldots, n,$$

where $\Pr(y_i = 1 \mid \boldsymbol{x}_i)$ is the probability that y_i is 1 given $\boldsymbol{x}_i$. $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^T$ is a p-dimensional regression coefficient vector and β_0 is an intercept. By using data-augmentation method with Pólya-Gamma distribution (Polson et al., 2013), the logistic regression model with Bayesian fused lasso (Kyung et al., 2010) is expressed as

$$\begin{aligned}
y_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0, w_i &\sim \mathrm{Binom}\left(1, \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta})}}\right), \\
w_i &\sim \mathrm{PG}(1, 0), \\
\beta_i \mid \lambda_1 &\sim \mathrm{Laplace}(\lambda_1), \\
\beta_i - \beta_j \mid \lambda_2 &\sim \mathrm{Laplace}(\lambda_2),
\end{aligned}$$

where $\mathrm{Laplace}(x \mid \lambda)$ is a Laplace distribution and $\mathrm{PG}(x \mid a, b)$ is a Pólya-Gamma distribution.

3 Proposed method

We propose the logistic regression model assuming a Laplace prior on regression coefficients and horseshoe prior (Carvalho et al., 2010) on the successive differences as follows:

$$\begin{aligned}
\beta_j \mid \tilde{\lambda}_1 &\sim \mathrm{Laplace}(\tilde{\lambda}_1), \\
\beta_j - \beta_{j-1} \mid \gamma_j^2, \tau^2 &\sim \mathrm{N}(0, \gamma_j^2 \tau^2), \\
\gamma_j &\sim \mathrm{C}^+(0, 1), \\
\tau &\sim \mathrm{C}^+(0, 1),
\end{aligned}$$

where $\mathrm{C}^+(0, a)$ $(a > 0)$ is a half-Cauchy distribution. Horseshoe prior has a heavy tail and an infinite spike at zero. Therefore, the large differences of regression coefficients will be remained and small differences are shrunk strongly toward zero in the proposed method.

With the hierarchical expression of Laplace prior (Andrews and Mallows, 1974) and the half-Cauchy distribution (Wand et al., 2011), the Gibbs sampler can be constructed.

4 Simulation studies

We conducted the Monte Carlo simulations with artificial data generated from the true model:

$$\Pr(y_i = 1 \mid \boldsymbol{x}_i) = \frac{1}{1 + e^{-(\beta_0^* + \boldsymbol{x}_i^T \boldsymbol{\beta}^*)}},$$

where β_0^* is a true intercept and $\boldsymbol{\beta}^*$ is a true p-dimensional regression coefficient vector. In addition, $\boldsymbol{x}_i$ $(i = 1, \ldots, n)$ is multivariate normally distributed as $\mathrm{N}(\mathbf{0}_p, \boldsymbol{\Sigma})$. Here, the diagonal element of Σ is 1 and the non-diagonal element is ρ.

We considered $\boldsymbol{\beta}^* = \boldsymbol{\beta}_1^*, \boldsymbol{\beta}_2^*$, where $\boldsymbol{\beta}_1^* = (\mathbf{1.0}_5^T, \mathbf{0.0}_5^T, \mathbf{1.0}_5^T, \mathbf{0.0}_5^T)^T$ and $\boldsymbol{\beta}_2^* = (-\mathbf{1.0}_5^T, \mathbf{0.0}_5^T, \mathbf{1.0}_5^T, \mathbf{0.0}_5^T)^T$. We simulated 100 datasets for each case.

We compared the proposed method (LBFH) with the logistic regression model with fused lasso (LFL), and the logistic regression model with Bayesian fused lasso (LBFL). To evaluate the accuracy of the estimation of regression coefficients, we computed mean squared error (MSE). To evaluate the accuracy of the estimation of successive differences of regression coefficients, we computed $\mathrm{MSE}_{\mathrm{diff}}$ as $\frac{1}{100}\sum_{k=1}^{100}\sum_{j=2}^{p}\left\{(\hat{\beta}_j^{(k)} - \hat{\beta}_{j-1}^{(k)}) - (\beta_j^* - \beta_{j-1}^*)\right\}^2$, where $\hat{\boldsymbol{\beta}}^{(k)} = (\hat{\beta}_1^{(k)}, \ldots, \hat{\beta}_p^{(k)})^T$ is the estimated values of regression coefficients for k-th dataset. To assess the prediction accuracy, we computed the negative expected log-likelihood (EL).

5 Results

From Table 1, we can find that LBFH gives the smallest values in $\mathrm{MSE}_{\mathrm{diff}}$ and EL for $\boldsymbol{\beta}_1^*$ and $\boldsymbol{\beta}_2^*$. When we compare Bayesian methods LBFL and LBFH, LBFH gives smaller MSE, $\mathrm{MSE}_{\mathrm{diff}}$ and EL for $\boldsymbol{\beta}_1^*$ and $\boldsymbol{\beta}_2^*$. On the whole, LBFH estimates the successive differences of regression coefficients more accurately and gives the better performance in prediction than the existing methods.

Table1: MSE (standard deviation), $\mathrm{MSE}_{\mathrm{diff}}$, and EL for $n = 100$ and $\rho = 0.5^{|i-j|}$.

		MSE (sd)	$\mathrm{MSE}_{\mathrm{diff}}$ (sd)	EL (sd)
$\boldsymbol{\beta}_1^*$	LFL	**1.633** (0.839)	2.176 (1.561)	31.328 (3.405)
	LBFL	3.270 (1.927)	4.856 (2.160)	34.972 (4.982)
	LBFH	1.724 (0.708)	**1.693** (0.539)	**31.096** (2.697)
$\boldsymbol{\beta}_2^*$	LFL	1.805 (1.058)	2.569 (2.586)	32.078 (3.566)
	LBFL	2.640 (1.286)	4.300 (2.201)	34.666 (3.875)
	LBFH	**1.768** (0.606)	**1.660** (0.467)	**31.380** (2.155)

6 Conclusion

In this paper, we propose the logistic regression model with Bayesian fused lasso with horseshoe prior. The results of the simulation studies show that the proposed method outperformed the existing method in the estimation accuracy of the successive differences of regression coefficients and the prediction accuracy.

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